

# thylinao at BAREC Shared Task 2026: Ensembling at the Noise Floor

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## Abstract

Team thylinao entered both tracks of the BAREC 2026 shared task on sentence-level Arabic readability with a deliberately plain system: a fourteen-member equal-weight ensemble of encoder fine-tunes, decoded to the 19-level scale by cutpoints interpolated halfway between validation-tuned thresholds and thresholds that match the blind score distribution to the training label prior. It scored 85.4 QWK on both tracks, second on both. This paper is about how those choices were measured. Three independent estimates place the noise floor of our offline evaluation between 0.09 and 0.44 QWK, as large as the margins that decided the final ranking. In four paired comparisons against the live board, only the smallest of the four offline estimates survived; the other three collapsed or reversed. We therefore proved that the Open and Strict tracks score identical gold, and made every decision worth a submission from a live board reading. When the board showed the mean was capped, we chose the final submission for maximum row-level difference from the incumbent, and it drew the closing 0.1. The full ledger, including five refuted hypotheses, is in the appendix.

## 1 Introduction

Midway through the BAREC 2026 campaign, our ensemble scored 85.44, 85.28 and 85.42 on the same test split while we varied nothing but the resolution of a threshold search grid. Chasing that anomaly produced the central finding of this paper: measured three independent ways, the noise floor of our offline evaluation sits between 0.09 and 0.44 QWK, while the margins that decided the final ranking were 0.1 to 0.4. An evaluation cannot choose between changes smaller than its own noise, and ours did not. In four paired tests, three of four offline estimates collapsed to zero or reversed in sign on the blind leaderboard. We responded by rebuilding the campaign around the

one instrument with adequate resolution, first proving that the Open and Strict tracks score identical blind gold, then making every decision worth a submission from a live board reading. This paper describes the system that finished second on both tracks at 85.4 QWK, and the measurement discipline that chose it, which we believe is the part worth reusing.<sup>1</sup>

The system is ordinary on purpose, and we report that plainness as a result rather than a limitation: every fitted alternative we tried, from non-negative least squares weights to per-member isotonic calibration and robust aggregators, scored below plain equal weight (Section 5). At our resolution, a more elaborate combiner could not be justified.

The first application of the rule came early. The public CAMEL-Lab readability checkpoints were fine-tuned on the BAREC corpus, which contains the shared task’s training split, so their training-split predictions are in-sample: solo QWK of 89 to 95 on train against 82 to 84 on clean holdouts. Any blend weight or cutpoint tuned on train would have been tuned on memorised labels, so we retired train as a tuning surface and adopted tune-on-validation, evaluate-once-on-test.

We contribute a noise-floor audit of offline evaluation on this task, with three independent estimators and four paired offline-versus-board tests on the true blind gold; a validated protocol for using a second track as a measurement instrument, including a byte-identical cross-track parity proof and two competition mechanics that changed our decisions more than any model change did; a cutpoint calibration that was the one lever whose effect cleared the noise floor, worth 84.1 to 84.5 on the board; and an audited ledger with five refuted hypotheses and a catalogue of improvements the

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<sup>1</sup>Code, the fine-tuning grid and the per-run score arrays: <https://github.com/thylinao1/barec-2026-readability>

ensemble absorbed without moving, so that neither is re-attempted (Appendices A and B).

## 2 Background

BAREC grades Arabic sentences on a 19-level ordinal scale aligned to a pedagogical reading progression (Elmadani et al., 2025b; Habash et al., 2025). The shared task evaluates systems with Quadratic Weighted Kappa (Elmadani et al., 2025a). QWK penalises an error by the square of its distance in levels, so a system is judged on the shape of its prediction distribution at least as much as on per-sentence accuracy. That drove most of what we did, and it is visible in the final standings: two teams finished with far higher exact accuracy than ours and far lower QWK.

We entered the sentence-level Open and Strict tracks. Three results from the 2025 edition shaped our plan. Ensembling across loss families was the winning margin: the winning system (Basem et al., 2025) combined four Arabic encoders under three loss families and reported 84.8 for its best single model rising to 87.5 for the full ensemble, and the runner-up stacked seven regression base learners under a meta-classifier (Saeed et al., 2025). Ordinal framing and D3Tok morphological tokenisation are table stakes, worth roughly +3.1 and +1.0 respectively on the organisers’ own baselines, and other entries explored ordinal heads, uncertainty-aware decoding and differentiable QWK surrogates (Nazzal, 2025; Abdou, 2025; Trigui, 2025; Mohamed Emad Elden, 2025). External data mostly did not help, and the organisers say so directly: the winning system scored worse in Open than in Strict. That last point is why our Open and Strict systems are identical. We did not treat the Open track as a licence to add outside corpora, but as a second measurement of the same blind gold, which turned out to be worth more.

## 3 System Overview

### 3.1 Members and blend

The final system averages fourteen z-standardised member scores at equal weight, standardising each member’s validation and blind scores independently before averaging. Four seats are public checkpoints, nine are seed-averaged families of our own fine-tunes and one is a single run; four use variants retrained on train plus the released test split, legal in both tracks. Members came from a cluster grid of 113 completed fine-tuning runs

over 13 backbones, 6 loss families and three input views, combined with public checkpoints into a pool of 172 candidates (Appendix D).

### 3.2 The measurement chain

Before training anything we reimplemented the official scorer and gated the project on exact agreement, reproducing the official script string-for-string on the development fixture including all six secondary metrics. A separate entry point pins the label set to [1, 19] for per-fold evaluation, where a sparse tail would otherwise silently build a smaller confusion matrix. The gate paid for itself in one upload: a submission whose local test QWK was 86.10 returned 86.10 on the board with all seven metrics matching, confirming that our grader is the board’s grader.

Preprocessing parity is a correctness concern of the same kind. The blind set consists of new sentences, so the corpus’s gold morphological views must be recomputed locally with CAMEL Tools (Obeid et al., 2020). Two failure modes here are silent: feeding raw text to a model that expects the Word view drops that member from 84 QWK to 50, and omitting only the character-normalisation step inside that pipeline drops exact match against gold from 99% to 31%. Neither raises an error; both produce a plausible submission.

The subtler version cost us a member. Our locally computed D3Tok matches the released gold at about 76%, because the released column used a license-gated morphological database. Measured naively this appeared to cost the regression member 84.68 to 82.75, so we shipped a word-only ensemble. That was wrong, and the error was not in the pipeline but in the standardisation: the comparison took the member’s validation scores from the gold column while taking its blind scores from ours. Those are different input distributions, and z-scoring cannot reconcile them. Recomputing validation through the same pipeline as blind made the member self-consistent, and adding it back was worth +0.3 on the board, more than every later calibration experiment combined. The rule: a member’s standardisation statistics and the scores they are applied to must come from one pipeline, end to end.

### 3.3 Half-match cutpoint calibration

Cutpoints tuned on validation put the blind predictions in the wrong place. The system predicted level 12 for about 8% of blind sentences where the

corpus prior puts roughly 21%. The scores shift between validation and blind, so cutpoints fitted in validation score space land wrongly in blind score space, and quadratic weighting punishes the resulting systematic level shift.

The correction uses the only label-free target available, the training label histogram. Let  $r$  be the validation-tuned cutpoint vector,  $p$  the training label prior over levels 1 to 19, and  $s_b$  the blind blended scores. We ship

$$c = \text{sort}((1 - f)r + fQ(s_b, \text{cumsum}(p)_{1:18})) \quad (1)$$

with  $f = 0.50$ , where  $Q$  takes empirical quantiles of  $s_b$  at the cumulative prior boundaries. The shipped cutpoints therefore sit halfway between what maximised QWK on validation and what would make the blind prediction histogram match the corpus prior exactly.

We measured  $f$  on the live board, because blind has no labels and the test split does not exhibit the shift that motivates the correction. The curve is single-peaked:  $f = 0$  gives 84.1,  $f = 0.50$  gives 84.5,  $f = 0.60$  gives 84.4, and  $f = 1$  gives 84.0. Both endpoints are worse than the midpoint, by 0.4 and 0.5, because the two targets carry complementary and partly wrong information: the rounder knows the score-to-level mapping but assumes the wrong score distribution, while the prior match knows the right output distribution but discards the rounder’s ordering information. After five readings we stopped tuning  $f$ .

## 4 Experimental Setup

The release provides train (54,845 sentences over 1,518 documents), validation (7,310) and test (7,286) with gold labels, and a blind test of 8,077 sentences over 212 documents. Folds are built once with grouped stratified 5-fold cross-validation, grouped on document so topic and vocabulary do not leak, and serialised to a committed fold map that is never regenerated. Our go/no-go number is a nested out-of-fold QWK in which cutpoints are fitted on the other folds and applied to the held-out fold, so thresholds never see the rows they score. Every member shows validation below test by about 1.3 QWK, which correctly predicted the blind set would score nearer 85 than 86.

**Calibrating the instrument.** We estimated the noise of that protocol three ways. Holding data,

members and weights fixed and varying only the threshold search grid between 150 and 500 candidates moves the same blend over a range of 0.345 QWK with standard deviation 0.094. Freezing the cutpoints and bootstrap-resampling the 7,286 test rows 400 times puts the sampling standard deviation of a single test measurement at about 0.44 QWK. Varying the fold-map seed, by contrast, returns bit-identical results, because the rounder snaps cutpoints to a fixed grid and small changes in the fitting rows often do not move the selected grid point at all. That third result is a trap worth naming: anyone estimating variance by reshuffling folds on a grid-search rounder will conclude, incorrectly, that their procedure is deterministic. Vary the grid instead. A paired bootstrap, with cutpoints frozen and the same resampled rows used for both systems, shrinks a naive +0.52 gain to +0.24 with a 95% interval of [+0.03, +0.45].

**The second track as an instrument.** Two byte-identical files submitted to Open and to Strict returned identical metric vectors, so both tracks score the same 8,077-row blind gold. This makes the Open board a free measurement instrument for a Strict system whenever the Open runner-up is far enough behind that a losing probe costs no place. Two mechanics constrain the protocol, and both cost us before we understood them. First, we believed Codabench displays each participant’s latest submission; verification against the competition API showed a submission rule of Add, meaning the board shows the entry the participant explicitly added, not the most recent upload, so probing is close to free. Second, ties break against the later submission, so only a strictly better score should ever be sent to a track we lead. We record both because a system description that omits them would misrepresent how the final submission was chosen.

## 5 Results

### 5.1 Offline estimates against the board

Table 1 pairs every offline estimate we acted on with the live board reading of the same change. Across four paired comparisons on the blind gold, three offline estimates collapsed to zero or reversed in sign on the board, and the only change that delivered was the one with the smallest offline estimate of the four.

The precision of an offline protocol is not its accuracy. Our nested out-of-fold measurements

| change                          | offline | board |
|---------------------------------|---------|-------|
| add MARBERTv2 (test holdout)    | +0.41   | 0.00  |
| add MARBERTv2 (nested OOF)      | +0.406  | -0.1  |
| document shrinkage $\alpha=0.1$ | +0.22   | -0.2  |
| add D3Tok regression member     | +0.086  | +0.3  |

Table 1: Paired offline-versus-board comparisons. Each row is one change, estimated offline and then measured on the blind gold.

carry a standard deviation of 0.019 across grid settings, which invites reading the +0.406 MARBERTv2 gain as twenty standard deviations of signal; the board returned  $-0.1$ . That 0.019 is the reproducibility of the procedure, not the uncertainty of the quantity it estimates. Nested cross-validation controls threshold overfitting and says nothing about distribution shift between the tuning and target samples, which is what separated validation from blind here.

## 5.2 Saturation

Once the ensemble reached fourteen members it absorbed everything. Against an equal-weight baseline of 85.281 on honest nested out-of-fold, twenty fitted-aggregation configurations scored between 84.74 and 85.12: non-negative least squares weights and ridge-shrunk variants, direct-QWK coordinate ascent, an expanded 44-member pool, per-member isotonic calibration, and median, trimmed-mean and rank-mean aggregation (Appendix C). Every one is below equal weight. Validation cannot support fourteen fitted parameters, and the row-wise extremes that robust aggregators discard carry real ordinal signal.

Member-level gains were absorbed too. A word-to-D3Tok lexicon raised sentence-exact fidelity from 63.7% to 79.1% and lifted the affected member by +0.91 QWK solo, yet moved the blend by between  $-0.02$  and  $+0.001$ : the cleaned member correlates more with the pack, so its solo gain trades one for one against lost diversity. Retraining the five strongest families on train plus test, 13% more gold data, moved the blend by +0.008.

## 5.3 The endgame

With the mean provably capped, we changed the selection criterion. A dual gate, requiring a candidate to beat the baseline on both honest nested out-of-fold and a deployment-mirrored oracle on a second labelled holdout, called four staged candidates statistical ties; probed on the board they

returned 85.2, 85.2, 85.2 and 85.3, four for four at the board’s resolution. So we stopped ranking candidates by offline score and ranked them by disagreement with the incumbent. Five near-clones, differing on 2% to 6% of rows, drew ties; the candidate that changed 811 of 8,077 predictions, 10.0%, drew the 0.1 that closed the campaign and became our final 85.4 on both tracks.

## 5.4 Final standing

We finished second on both tracks with 85.4 QWK, 37.8 exact accuracy, 71.3 adjacent accuracy, average distance 1.1, and 60.1 / 67.1 / 74.8 at 7, 5 and 3 collapsed levels. Open ranked 4 submissions and the winner scored 85.5; Strict ranked 11, and the winner also scored 85.4, taking first on the earlier timestamp. The winner’s exact accuracy exceeded ours on both tracks, 39.5 against 37.8 in Open, marking a system-level difference our pool could not reach by recombination, calibration or retraining.

## 6 Conclusion

Three conclusions, in ascending order of how uncomfortable they are. First, a nested out-of-fold estimate is not automatically an honest estimate; ours was correctly specified and still overpromised by 0.5 QWK. Second, the reported standard deviation of a repeated measurement is not the uncertainty of what it measures. Third, on this task the leaderboard was the only instrument with adequate resolution, and the second track let us use it without risk. That is not general advice, since it requires two tracks scoring the same gold and a margin wide enough to make a losing probe free; teams without one should budget submissions as measurements rather than as attempts.

The counterfactual is worth stating plainly. Our own prioritisation document, written before these probes, ranked recovering the MARBERTv2 weights as the highest-value action remaining, on a +0.41 offline estimate. Following it as written would have shipped 84.7. If a seven-thousand-row holdout cannot resolve 0.2 QWK, then some fraction of the ablation gains published for QWK-scored tasks, possibly including some in this paper, are reports of noise. The useful response is for participants with board access to publish paired offline-versus-board comparisons routinely, so the field can estimate how much of its reported progress survives a fresh sample.

## Limitations

Our noise-floor evidence rests on four paired comparisons, which shows that the disagreement occurs but not how often. Board readings are single measurements on a fixed 8,077-row set and carry their own sampling noise, which we could not estimate because we cannot resample the blind gold. The second-track oracle depends on a margin over the runner-up and was unavailable to us once that margin closed. Our locally computed D3Tok is systematically degraded relative to a licensed team's, and the standardisation fix recovers the member's usefulness without recovering that loss. Finally, equal weights and a single calibration scalar are unrefuted at our measurement resolution rather than shown optimal; we cannot rule out that a better combiner exists below our noise floor.

## Acknowledgements

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## A Submission ledger

Every row is a live board reading on the blind gold. Rows marked with a dagger were probes uploaded to the Open track and read but never added to a leaderboard, which the competition’s Add submission rule makes free (Section 4).

| system                                   | calibration | QWK          |
|------------------------------------------|-------------|--------------|
| word-only 3-member                       | $f = 0$     | 84.1         |
| <b>half-match champion</b>               | $f = 0.50$  | <b>84.5</b>  |
| match-60                                 | $f = 0.60$  | 84.4         |
| full distribution match                  | $f = 1.00$  | 84.0         |
| metadata-reweighted prior <sup>†</sup>   | $f = 0.50$  | 83.9         |
| <b>blend4reg</b> (both tracks)           | $f = 0.50$  | <b>84.8</b>  |
| blend5, adds MARBERTv2 <sup>†</sup>      | $f = 0.50$  | 84.7         |
| blend4reg + doc shrinkage <sup>†</sup>   | $f = 0.50$  | 84.6         |
| fleet11                                  | $f = 0.50$  | 85.1         |
| fleet10strict                            | $f = 0.50$  | 85.2         |
| fleet14 (both tracks)                    | $f = 0.50$  | 85.3         |
| fleet14 tie probes, 4 files <sup>†</sup> | $f = 0.50$  | 85.2 to 85.3 |
| <b>fleet14tt4</b> , final, both          | $f = 0.50$  | <b>85.4</b>  |

Table 2: Blind-phase submissions, in the order they were measured.

Two rows in our internal log recorded a track that we could not re-confirm after the phase closed, and one calibration reading at  $f = 0.40$  came from an ambiguous screenshot. The authenticated submission history is not reachable through the public API, so rather than guess we dropped the  $f = 0.40$  point and describe the calibration peak as flat between 0.5 and 0.6 rather than across 0.4 to 0.6.

## B Negative results in full

**Metadata-reweighted label prior, refuted at 83.9.** The half-match calibration matches the blind scores to the raw training label prior. That target is arguably wrong, because the blind set’s composition differs sharply from train: the STEM domain rises from 6.9% to 17.8% and the Specialized text class from 26.9% to 44.7%, while Advanced falls from 33.0% to 19.3%. Domain and text class are present for 100% of blind rows and take values that exist in train, which permits a label-free reweighting  $P(\ell) = \sum_s P(s \mid \text{blind}) P(\ell \mid s, \text{train})$  over  $s = \text{Domain} \times \text{Text\_Class}$ . This shifts the prior by  $L_1$  distance 0.1224 and alters 1,697 of 8,077 predictions, every one by exactly +1 level. We built it as a strictly single-variable change: the implementation’s own raw-prior reference output is byte-identical to the previously submitted champion, so the prior is provably the only thing that differs. It scored 83.9,

losing 0.6.

The hypothesis could not be tested offline, because blind has no labels and the test split does not exhibit the shift that motivated the correction. What the refutation rules out is specific: pushing 21% of predictions up one level makes things worse, so the blind set is not systematically harder in readability level than train despite being 2.6 times more STEM. We suspect this is because BAREC levels are annotated against a reading progression rather than topic difficulty, so domain composition and level composition are less coupled than intuition suggests, but we did not test that explanation.

**MARBERTv2 as a member, measured at 0.00 and  $-0.1$ .** MARBERTv2 was the most attractive remaining candidate on paper: a different backbone family, and the least correlated with our blend at 0.945 against 0.96 to 0.98 for everything else. Offline it measured  $+0.41$  on the clean test holdout and  $+0.406$  on honest nested out-of-fold, the largest gain in either table. Recovering it was expensive, because the original fine-tuning script saved only prediction arrays and never called `save_pretrained`, so the model existed only as numbers and had to be retrained. On the board it contributed 0.00 when added to the champion and  $-0.1$  in the five-member configuration.

**Document-level score shrinkage, measured at  $-0.2$ .** The blind set ships a document column with 212 documents over 8,077 sentences, which invites smoothing each sentence score toward its document mean,  $s \leftarrow (1-\alpha)s + \alpha \text{mean}_{\text{doc}}(s)$ . On the clean test holdout the effect is single-peaked and sharp: 85.48 at  $\alpha = 0$ , 85.69 at 0.1, 85.32 at 0.2, 84.88 at 0.3, and a collapse to 82.15 at 0.5. We shipped the peak and it cost 0.2. One implementation subtlety has to be right for the transform to be the one that was measured: shrinkage must be applied to both validation and blind scores, because the rounder’s cutpoints live in validation score space.

**Weak or redundant members.** AraELECTRA regression is the most decorrelated candidate we measured at 0.9101, but at 78.29 solo it is too weak to help and moves the blend from 85.07 to 84.87. A D3Tok cross-entropy member costs  $-0.055$  on honest nested out-of-fold and was excluded even though its sibling regression member is our best

single addition. Decorrelation alone does not qualify a member; our working rule is that a member must be either near 85 solo or below 0.90 correlated, and one that is neither is not worth a seat.

## C The saturation sweep

All figures are honest validation nested out-of-fold QWK, mean over threshold grids of 200, 300 and 400 candidates. The equal-weight fourteen-member baseline is  $85.281 \pm 0.051$ .

| aggregation                         | QWK            |
|-------------------------------------|----------------|
| <b>equal weight (shipped)</b>       | <b>85.281</b>  |
| NNLS weights, ridge-shrunk 25/50/75 | 84.79 to 85.10 |
| OLS and clipped weights             | 84.79 to 85.10 |
| direct-QWK coordinate ascent        | 84.97, 84.98   |
| expanded 44-member pool, NNLS       | 84.94 to 85.10 |
| per-member isotonic calibration     | 84.74 to 85.12 |
| median / trim1 / trim2 / rank-mean  | 84.79 to 85.01 |

Table 3: Twenty aggregation configurations, all below plain equal weight. Per-fold fitted weights were unstable, zeroing members in some folds and not others.

Member-level improvements were absorbed as well. We built a word-to-D3Tok lexicon from the gold corpus alignment (98.8% of sentences align), covering 92% of validation tokens and 91% of blind tokens, with the older disambiguation pipeline filling the remainder. This raised sentence-exact fidelity against the gold column from 63.7% to 79.1% and lifted the affected regression member from 82.27 to 83.18 solo on validation, recovering most of the gap to gold text. At blend level the same change was worth  $-0.019$  when swapped for the single member,  $+0.001$  for the seed-averaged family, and  $-0.024$  for both together. Retraining the five strongest families on train plus the released test split, 62,131 rows and a 13% increase in gold data, moved the blend from 85.281 to 85.289.

## D Fine-tuning grid

Both tracks we entered are satisfied by BAREC data alone. We used no external readability resources: the SAMER corpus (Alhafni et al., 2024) and the SAMER readability lexicon (Al Khalil et al., 2020), which the Constrained track permits, were not used, and we added nothing outside BAREC in the Open track either, since the 2025 evidence is that external data does not help on this task.

Training used a Slurm job array on A100 GPUs. One trainer script takes backbone, loss, input view,

| origin                         | heads / inputs            | seats |
|--------------------------------|---------------------------|-------|
| public checkpoints             | CE, reg; Word, D3Tok, raw | 4     |
| own AraBERT v02/v2/large       | CE, reg; Word, D3Tok      | 5     |
| own CAMeLBERT, QARiB           | CE, reg; Word, raw        | 3     |
| own MARBERT, AraELECTRA, XLM-R | CE, reg; raw, Word        | 3     |

Table 4: The fourteen seats, by origin. Nine are seed-averaged families of our own fine-tunes, four are public checkpoints, one is a single run; four use variants re-trained on train plus the released test split, legal in both tracks. The exact member list is given below.

seed, learning rate and warmup ratio, and writes validation and test score arrays through the same code path used at inference. Defaults: learning rate  $2 \times 10^{-5}$ , 5 epochs, batch 16, maximum sequence length 128, no warmup except 0.06 for the large backbones. Cross-entropy, soft-label and focal heads are decoded to a continuous score by the posterior mean  $\text{softmax}(z) \cdot [1..19]$ ; regression heads use the raw output; CORN follows Shi et al. (2023).

The dispatched grid has 112 rows over 13 backbones and 6 loss families (cross-entropy 45, regression 44, soft-QWK 10, focal 6, CORN 5, soft-label 2), crossing three input views (raw, Word, D3Tok) with seeds. 113 runs completed with a done marker and both score arrays. With public checkpoints and seed-averaged families this produced a pool of 172 candidate members, 26 of them seed averages.

Two gates apply before a member may enter a blend, both added after an incident in which a checkpoint export was omitted and only prediction arrays survived. The head shape must match the fine-tune rather than the base model, and locally recomputed validation scores must reproduce the cluster’s saved arrays above 0.99 correlation. A member failing either gate is not blended, because a standardisation computed from mismatched scores is silently wrong rather than loudly wrong.

The fourteen seats of the final system are: the four public-checkpoint champion members (arabertv02-word-CE, camelbert-word-CE, qarib-barec-optimized-v8, arabertv2-d3tok-reg scored through our own D3Tok pipeline); seed-averaged families of arabertv2-reg-d3tok, arabertv02-ce-word, alarge02-reg-word and marbert-ce-raw, each retrained on train plus test; seed-averaged families of camelbert-ce-word, qarib-ce-raw,

araelectra-reg-word, camelbert-reg-word and arabertv2-ce-d3tok; and a single xlm-roberta-large regression run.

## E Harness detail

The rounder maps a continuous score to a level in [1, 19] via 18 monotone cutpoints chosen by coordinate ascent to maximise QWK. The search is seeded from equal-frequency quantiles of the label distribution, each cutpoint is swept over a fixed grid spanning the score range, monotonicity is enforced between neighbours, and the output is clipped. We use monotone coordinate ascent rather than Nelder-Mead, which can return crossed, non-monotone thresholds in 18 dimensions and is retained only as a cross-check. On a classical ridge baseline the protocol reads cleanly: naive rounding 71.25, nested out-of-fold 74.48, non-nested out-of-fold 74.63.

Folds are built once with grouped stratified 5-fold cross-validation, grouped on document and stratified on the 19-level label, and serialised to a committed document-to-fold map covering 1,518 documents. The map is loaded identically on every machine and never regenerated, because regenerating it per machine is how out-of-fold arrays silently fail to line up. All five folds carry all 19 levels.

The parity gate compares our scorer against the official development fixture at the level of the printed string: QWK 80.0040%, accuracy 56.6211%, adjacent accuracy 69.8632%, average absolute distance 1.143776, and 65.8687% / 70.2736% / 76.4569% at 7, 5 and 3 collapsed levels.

## F Reproducibility

Code, the fine-tuning grid, and the per-run validation and test score arrays are at <https://github.com/thylinao1/barec-2026-readability> under an MIT license. The final submission is reproduced from that repository byte for byte; its SHA-256 is e243ab46a0edb09e1361e8bcb52159708e2f22bb4d90063c5206932db0bf2516, and the exact command is in the repository README.

Predictions over the blind test set are deliberately not published. The blind set is released only to registered participants and may be reused in a future edition, so a strong prediction file over it would function as a pseudo-label source for later

participants. The code regenerates those files for anyone holding the blind input, and the hash above lets the organisers verify the submitted file exactly.